

In this practical machine learning lab, I built a full classification pipeline using the Scikit-Learn Wine dataset. First, I got the setup ready. Then I looked at the data with basic Exploratory Data Analysis. I also used heatmap plots to see patterns. Next, I split the dataset into training and test parts. I used stratification so the class mix stayed similar in both sets. After that, I trained two models: Logistic Regression and a Decision Tree. I checked how well they worked using accuracy scores. I also reviewed classification reports and confusion matrices. To learn more, I tried different feature choices. For example, I focused on alcohol, color_intensity, and proline to see how each one affected prediction strength. At the end, I answered a concept question about learning types. I sorted examples into supervised, unsupervised, and reinforcement learning.